Collaborative Annotation as an Interface Metaphor for Personalizable AI Reading Support

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Abstract

In this position paper, we argue that collaborative annotation is a compelling interface metaphor through which to empower end-users to develop personalized AI reading support in the form of in-document highlights. Collaborative annotation confers many of the same benefits of traditional data annotation tools for specifying model behavior while not removing users from the context of the original document. This paradigm preserves the benefits of existing inline document markup tools, as well as the contextual, interpretive character of reading, while incorporating the flexibility and power of LLM-based annotation. Through what we call *collaborative machine annotation*, users can iteratively develop models as they refine their own mental models, and use the deep understanding of that model's behavior for wider corpus analysis.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; *Web-based interaction*; • **Computing methodologies** → **Information extraction**.

Keywords

Collaborative Annotation, Personalized Reading Support, Information Extraction

1 Introduction

Reading assistance tools have long made use of AI methods to provide in-context augmentation of document content. Tools like Scim [9] and PaperPlain [3] make use of machine learning to augment paper content to demonstrate a paper's rhetorical structure, direct a reader's focus, provide definitions, and more. These types of tools that provide in-context augmentation of paper contents are typically tightly scoped to an intended context of use and audience, and are therefore able to make assumptions about those contexts to deliver specific and reliable information; because the information is in-context on the document, it is also typically easily verifiable. However, assumptions about usage context and audience come at the cost of generality, even in the domains in which the tools are deployed. Scim highlights, for instance, generalize poorly outside of the experimental sciences, even as the Semantic Reader[18], in which Scim is deployed, makes them available on a much wider range of academic papers. Further, because these tools are based on models that rely on static schemas, they cannot account for readers' preferences and idiosyncrasies in reading support; readers often rely on individually specific cues to make sense of papers [12].

By contrast, tools based on large language models (LLMs), whose behavior can be specified by users at inference time has led to the proliferation of systems like SemanticScholar's "ask this PDF" feature, or modes in popular LLM-based chatbots that allow uploaded documents that allow users to "chat" with the contents of a document. These systems are extremely flexible, and allow users to express complex and idiosyncratic information needs in natural language, and will sometimes anchor their responses to specific locations within documents. However, these chat-based tools introduce their own issues. While they allow end-user behavior specification and therefore personalized responses, verifying the reliability of the generated responses can be challenging. LLMs often exhibit a jagged capability frontier, where estimating where a model might succeed and fail is not always intuitive, and validating generated text is significantly more difficult to verify than in-context highlights. Additionally, these tools allow the specification of their behavior only in text, and keep no record of user preferences or reading focuses in between documents.

This paper argues that collaborative annotation—as exemplified by tools like Hypothesis [1] and Margin [2]—can provide an intuitive interface metaphor that allows for the flexibility of dynamic, user-specified LLM-based reading assistance, while emphasizing the user agency in evaluation, verifiability and context-awareness that inline highlighting tools can provide. Collaborative annotation between humans, in addition to its studied benefits for document comprehension, can provide in-context views of data to be annotated for granular specification of model behavior, and treating model suggestions as another collaborator can reduce the overhead for evaluating model performance and improving it with techniques like active learning. In-context experience with model suggestions also allows for deep, qualitative understanding of model performance such that users can make informed choices about model utility, differential use of model functions, and deploying the model to wider contexts, such as corpus-level annotation.

2 Collaborative Annotation as an Interface Metaphor

Collaborative annotation is a highly overloaded term in computing. It can refer to a variety of different processes, from the reconciliation of annotations in interfaces purpose-built for annotation such as Inception [17] or BRAT [24], to collaborative markup of educational materials as part of classroom assignments [20, 27]. In this paper, we focus primarily on the latter kind of collaborative annotation, and specifically tools like Hypothesis that enable collaborative annotation on any web content. We argue that this focus on in-context annotation allows for workflows similar to those in dedicated annotation tools, while presenting significantly less overhead than standalone annotation tools for web data.